APM5557: Partial Differential Equations
Homework Problems

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Problem 1. From page 40 in the notes.

for the PDE: $u_t - c^2 u_{xx} = 0$, with initial condition $u(x,0) = g(x)$, show that the solution is given by the formula:

$$u(x,t) = \frac{1}{2c\sqrt{\pi t}} \int_{-\infty}^{\infty} e^{-\frac{(x-\xi)^2}{4c^2 t}} g(\xi) d\xi \tag{1}$$

Solution. First compute the time derivative of u. Bringing $\frac{\partial}{\partial t}$ inside the integral gives

$$u_t = \int_{-\infty}^{\infty} \frac{\partial}{\partial t} \left[\frac{1}{2c\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \tag{2}$$

Applying the product rule to the terms in brackets,

$$u_t = \int_{-\infty}^{\infty} \left[-\frac{1}{4c\sqrt{\pi}\, t^{3/2}} e^{-\frac{(x-\xi)^2}{4c^2 t}} + \frac{1}{2c\sqrt{\pi t}} \cdot \frac{(x-\xi)^2}{4c^2 t^2} e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \tag{3}$$

Next compute the spatial derivative of u in the same way. Bringing $\frac{\partial}{\partial x}$ inside the integral gives

$$u_x = \int_{-\infty}^{\infty} \frac{\partial}{\partial x} \left[\frac{1}{2c\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \tag{4}$$

Since the prefactor does not depend on x, only the exponential is differentiated, so

$$u_x = \int_{-\infty}^{\infty} \frac{1}{2c\sqrt{\pi t}} \cdot \left(-\frac{x-\xi}{2c^2 t} \right) e^{-\frac{(x-\xi)^2}{4c^2 t}} g(\xi)\, d\xi. \tag{5}$$

Taking one more derivative of (5) with respect to x gives

$$u_{xx} = \int_{-\infty}^{\infty} \frac{1}{2c\sqrt{\pi t}} \frac{\partial}{\partial x} \left[-\frac{x-\xi}{2c^2 t} e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \tag{6}$$

Applying the product rule to the terms in brackets,

$$u_{xx} = \int_{-\infty}^{\infty} \frac{1}{2c\sqrt{\pi t}} \left[-\frac{1}{2c^2 t} e^{-\frac{(x-\xi)^2}{4c^2 t}} + \frac{(x-\xi)^2}{4c^4 t^2} e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \tag{7}$$

Multiplying (7) by c^2 cancels some of the factors of c, so

$$c^2 u_{xx} = \int_{-\infty}^{\infty} \frac{1}{2c\sqrt{\pi t}} \left[-\frac{1}{2t} e^{-\frac{(x-\xi)^2}{4c^2 t}} + \frac{(x-\xi)^2}{4c^2 t^2} e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \tag{8}$$

Substituting (3) and (8) into the PDE gives

$$\begin{aligned} u_t - c^2 u_{xx} = &\int_{-\infty}^{\infty} \left[-\frac{1}{4c\sqrt{\pi}\, t^{3/2}}\, e^{-\frac{(x-\xi)^2}{4c^2 t}} + \frac{1}{2c\sqrt{\pi t}} \cdot \frac{(x-\xi)^2}{4c^2 t^2}\, e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi \\ &- \int_{-\infty}^{\infty} \frac{1}{2c\sqrt{\pi t}} \left[-\frac{1}{2t}\, e^{-\frac{(x-\xi)^2}{4c^2 t}} + \frac{(x-\xi)^2}{4c^2 t^2}\, e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \end{aligned} \tag{9}$$

Since

$$\frac{1}{2c\sqrt{\pi t}} \cdot \frac{1}{2t} = \frac{1}{4c\sqrt{\pi}\, t^{3/2}}, \tag{10}$$

the two integrals in (9) have identical integrands, so everything cancels and

$$u_t - c^2 u_{xx} = 0. \tag{11}$$

□

Problem 2. From 3.3-1 in the book.

Let Ω be a smooth bounded domain in $\mathbb{R}^n$. For a C^2 solution of u of the wave equation $u_{tt} = c^2 \Delta u$, show that if the energy is defined as $E_\Omega(t) = \int_\Omega (u_t^2 + c^2 |\nabla u|^2) dx$, and either $u(x,t) = 0$ or $\frac{\partial u}{\partial n} = 0$ on $\partial\Omega$, then $E_\Omega(t)$ is constant in time.

Solution. To show that $E_\Omega(t)$ is constant, take its time derivative and set it equal to 0:

$$\frac{dE_\Omega}{dt} = \frac{d}{dt}\, \frac{1}{2} \int_\Omega \left(u_t^2 + c^2 |\nabla u|^2 \right) dx = 0. \tag{12}$$

Splitting this into two integrals,

$$\frac{d}{dt}\, \frac{1}{2} \int_\Omega u_t^2\, dx + \frac{d}{dt}\, \frac{1}{2} \int_\Omega c^2 |\nabla u|^2\, dx = 0. \tag{13}$$

Bringing $\frac{d}{dt}$ inside the first integral, u_t^2 becomes $2u_{tt}u_t$, which cancels the $\frac{1}{2}$ for that term:

$$\int_\Omega u_{tt} u_t\, dx + \frac{d}{dt}\, \frac{1}{2} \int_\Omega c^2 |\nabla u|^2\, dx = 0. \tag{14}$$

Bringing $\frac{d}{dt}$ inside the second integral and writing $|\nabla u|^2 = \sum_{i=1}^n u_{x_i}^2$, each $u_{x_i}^2$ becomes $2u_{x_i}u_{x_i t}$, which cancels the $\frac{1}{2}$ for that term. Since c is constant, we can also pull c^2 out of the integral:

$$\int_\Omega u_{tt} u_t\, dx + c^2 \int_\Omega \sum_{i=1}^n u_{x_i} u_{x_i t}\, dx = 0. \tag{15}$$

Next we show that the second integral in (15) is equivalent to the integral of $-u_t \Delta u$, by integrating by parts once in each x_i. Since $u_{x_i t} = (u_t)_{x_i}$, each term of the sum is $u_{x_i}(u_t)_{x_i}$. Integrating by parts in x_i moves the derivative off of u_t and onto u_{x_i}. By the divergence theorem, with n_i the i-th component of the outward normal,

$$\int_\Omega u_{x_i} (u_t)_{x_i}\, dx = \int_{\partial\Omega} u_t\, u_{x_i} n_i\, dS - \int_\Omega u_t\, u_{x_i x_i}\, dx. \tag{16}$$

Summing over i, and using $\sum_{i=1}^n u_{x_i} n_i = \nabla u \cdot n = \frac{\partial u}{\partial n}$ and $\sum_{i=1}^n u_{x_i x_i} = \Delta u$,

$$\int_\Omega \sum_{i=1}^n u_{x_i} u_{x_i t}\, dx = \int_{\partial\Omega} u_t\, \frac{\partial u}{\partial n}\, dS - \int_\Omega u_t \Delta u\, dx. \tag{17}$$

(This is Green's first identity with $f = u_t$ and $g = u$.) The boundary integral vanishes under either boundary condition. If $u = 0$ on $\partial\Omega$ for all t, then u does not change in time there, so $u_t = 0$ on $\partial\Omega$. If $\frac{\partial u}{\partial n} = 0$ on $\partial\Omega$, then that factor is zero directly. Therefore

$$\int_\Omega \sum_{i=1}^{n} u_{x_i} u_{x_i t}\, dx = -\int_\Omega u_t \Delta u\, dx. \tag{18}$$

Substituting (18) into (15),

$$\int_\Omega u_{tt} u_t\, dx - c^2 \int_\Omega u_t \Delta u\, dx = 0. \tag{19}$$

Combining the terms and factoring out u_t,

$$\frac{dE_\Omega}{dt} = \int_\Omega u_t \left(u_{tt} - c^2 \Delta u\right) dx = 0. \tag{20}$$

The second factor is the original PDE, which is 0. Since the change in energy is 0, $E_\Omega(t)$ is constant in time. □

Problem 3. From 3.3-2a in the book.

For the non-homogeneous wave equation $u_{tt} = \Delta u + f(x,t)$, show uniqueness of the solution u given the Dirichlet condition $u = g$ on $\partial\Omega$.

Proof. Assume we have two solutions u_1 and u_2 to the non-homogeneous wave equation, with the same boundary and initial conditions. Define $v = u_1 - u_2$. Then v satisfies the homogeneous wave equation:

$u_{1tt} = \Delta u_1 + f(x,t)$ in Ω
$u_{2tt} = \Delta u_2 + f(x,t)$ in Ω
$v_{tt} = \Delta v$ in Ω
$u_1 = g$ on $\partial\Omega$
$u_2 = g$ on $\partial\Omega$
$v = 0$ on $\partial\Omega$

By the same reasoning, we also have the initial conditions:

$v(x,0) = 0$ in Ω
$v_t(x,0) = 0$ in Ω

By the previous problem applied with $c = 1$, we know that the energy $E(t) = \int_\Omega (v_t^2 + |\nabla v|^2) dx$ is constant in time.

At $t = 0$, $v(x,0) = 0$ in Ω gives $\nabla v(x,0) = 0$, and $v_t(x,0) = 0$, so $E(0) = 0$. Since E is constant, $E(t) = 0$ for all t. The integrand $v_t^2 + |\nabla v|^2$ is continuous and nonnegative with zero integral, so $v_t = 0$ everywhere. Therefore $v(x,t) = v(x,0) = 0$, so $u_1 = u_2$. □

Problem 4. From 3.3-2b in the book.

For the non-homogeneous wave equation $u_{tt} = \Delta u + f(x,t)$, show uniqueness of the solution u given the Neumann condition $\frac{\partial u}{\partial n} = h$ on $\partial\Omega$.

Proof. Assume we have two solutions u_1 and u_2 to the non-homogeneous wave equation, with the same boundary and initial conditions. Define $v = u_1 - u_2$. Then v satisfies the homogeneous wave equation:

$u_{1tt} = \Delta u_1 + f(x,t)$ in Ω
$u_{2tt} = \Delta u_2 + f(x,t)$ in Ω
$v_{tt} = \Delta v$ in Ω

$\frac{\partial u_1}{\partial n} = h$ on $\partial\Omega$
$\frac{\partial u_2}{\partial n} = h$ on $\partial\Omega$
$\frac{\partial v}{\partial n} = 0$ on $\partial\Omega$
By the same reasoning, we also have the initial conditions:
$v(x,0) = 0$ in Ω
$v_t(x,0) = 0$ in Ω
By Problem 2 applied with $c = 1$, we know that the energy $E(t) = \int_\Omega (v_t^2 + |\nabla v|^2) dx$ is constant in time.

At $t = 0$, $v(x,0) = 0$ in Ω gives $\nabla v(x,0) = 0$, and $v_t(x,0) = 0$, so $E(0) = 0$. Since E is constant, $E(t) = 0$ for all t. The integrand $v_t^2 + |\nabla v|^2$ is continuous and nonnegative with zero integral, so $v_t = 0$ everywhere. Therefore $v(x,t) = v(x,0) = 0$, so $u_1 = u_2$. □

Problem 5. From 3.3-6 in the book.
We have a flexible beam with the ends clamped at $x = 0$ and $x = 1$.
We have the following:
$u_{tt} + \gamma^2 u_{xxxx} = 0$
$u(0,t) = 0$ and $u(1,t) = 0$ for all t
$u_x(0,t) = 0$ and $u_x(1,t) = 0$ for all t
Show that energy is conserved, where the energy is defined as:
$E = \frac{1}{2}\int_0^1 (u_t^2 + \gamma^2 u_{xx}^2) dx$.

Proof. This is similar to Problem 2, except that it is only in $\mathbb{R}^1$. To show that E is constant, take its time derivative and set it equal to 0:

$$\frac{dE}{dt} = \frac{d}{dt}\frac{1}{2}\int_0^1 \left(u_t^2 + \gamma^2 u_{xx}^2\right) dx = 0. \tag{21}$$

Splitting this into two integrals,

$$\frac{d}{dt}\frac{1}{2}\int_0^1 u_t^2\, dx + \frac{d}{dt}\frac{1}{2}\int_0^1 \gamma^2 u_{xx}^2\, dx = 0. \tag{22}$$

Bringing $\frac{d}{dt}$ inside the first integral, u_t^2 becomes $2u_t u_{tt}$, which cancels the $\frac{1}{2}$ for that term:

$$\int_0^1 u_t u_{tt}\, dx + \frac{d}{dt}\frac{1}{2}\int_0^1 \gamma^2 u_{xx}^2\, dx = 0. \tag{23}$$

Bringing $\frac{d}{dt}$ inside the second integral in the same way, u_{xx}^2 becomes $2u_{xx}u_{xxt}$, which cancels the $\frac{1}{2}$ for that term. Since γ is constant, we can also pull γ^2 out of the integral:

$$\int_0^1 u_t u_{tt}\, dx + \gamma^2 \int_0^1 u_{xx} u_{xxt}\, dx = 0. \tag{24}$$

Now use integration by parts twice on the second term. The first time,

$$\int_0^1 u_{xx} u_{xxt}\, dx = \Big[u_{xx} u_{xt}\Big]_0^1 - \int_0^1 u_{xxx} u_{xt}\, dx. \tag{25}$$

Since $u_x(0,t) = 0$ and $u_x(1,t) = 0$ for all t, also $u_{xt}(0,t) = 0$ and $u_{xt}(1,t) = 0$, so the boundary term is 0 and

$$\int_0^1 u_{xx} u_{xxt}\, dx = -\int_0^1 u_{xxx} u_{xt}\, dx. \tag{26}$$

The second time,

$$-\int_0^1 u_{xxx}u_{xt}\,dx = -\Big[u_{xxx}u_t\Big]_0^1 + \int_0^1 u_{xxxx}u_t\,dx. \tag{27}$$

Since $u(0,t) = 0$ and $u(1,t) = 0$ for all t, also $u_t(0,t) = 0$ and $u_t(1,t) = 0$, so this boundary term is also 0 and

$$\int_0^1 u_{xx}u_{xxt}\,dx = \int_0^1 u_{xxxx}u_t\,dx. \tag{28}$$

Substituting (28) back into (24),

$$\int_0^1 u_t u_{tt}\,dx + \gamma^2 \int_0^1 u_{xxxx}u_t\,dx = 0. \tag{29}$$

Combining the terms and factoring out u_t,

$$\frac{dE}{dt} = \int_0^1 u_t\left(u_{tt} + \gamma^2 u_{xxxx}\right)dx = 0. \tag{30}$$

The second factor is the original PDE, which is 0. Since the change in energy is 0, E is constant in time, so energy is conserved. □